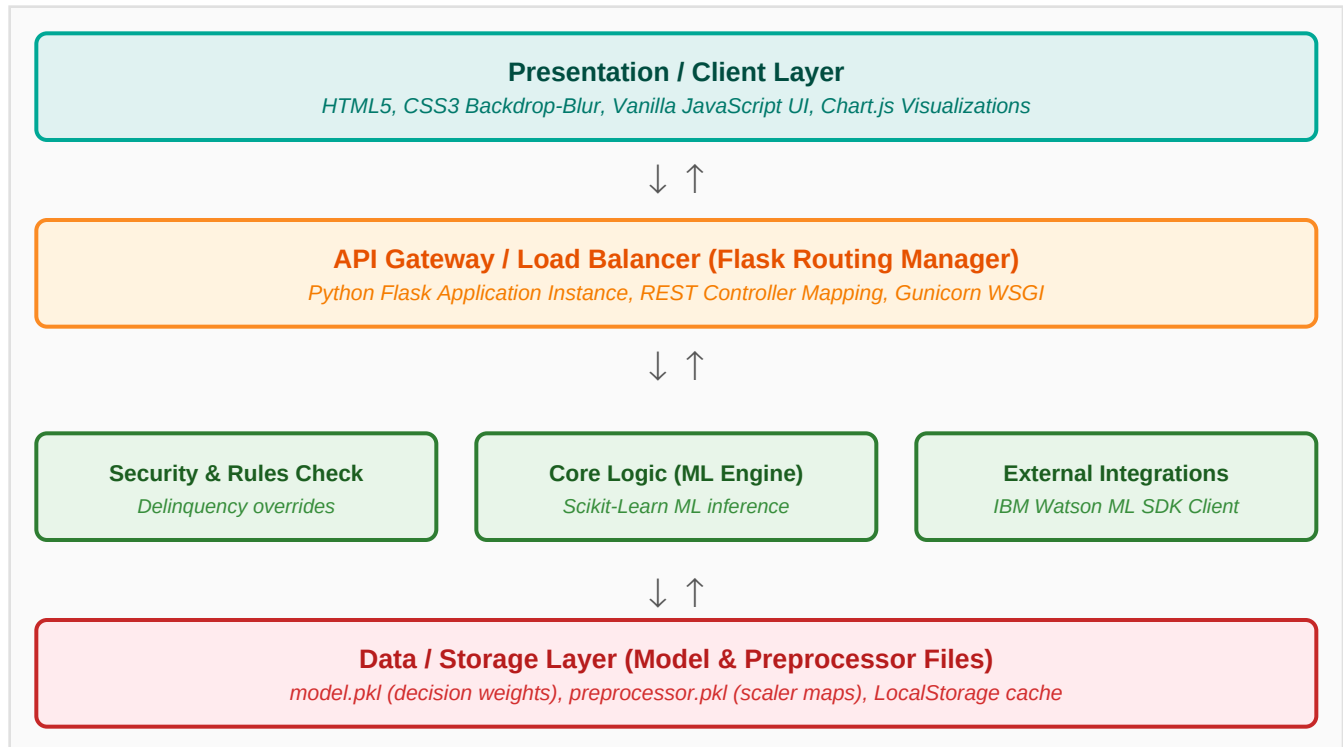


Solution Architecture

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the responsive dashboard interface, handles parameter user entries, plots model metric comparison matrices, outputs explainable dial probability score charts, and runs customer pre-qualification check wizards.	HTML5, Vanilla CSS3 (backdrop-blur glassmorphism), Vanilla JavaScript, Chart.js, FontAwesome
API Gateway	Acts as the front door application controller. Exposes REST API endpoint routes, directs JSON input data packets to modeling engines, handles CORS access, and manages HTTP request/response payloads.	Python Flask API Instance, Gunicorn WSGI Server
Microservices / Core App	Executes the data processing and machine learning inference pipelines. Preprocesses category mappings, scales numerical variables, queries decision boundaries using Decision Tree estimators, and extracts positive/negative risk contributors.	Scikit-Learn (sklearn), Pandas, NumPy, Python 3.12
Database / Storage Layer	Caches the serialized binary model configurations and weights, avoiding transactional SQL database lags. Stores pre-qualification scaling assets and manages browser-local history logs.	model.pkl (active classifier weights), preprocessor.pkl (scaler metrics), HTML5 LocalStorage